EXP05

CODE:

```
import matplotlib.pyplot as pltimport numpy as np
# Simulated training results (demonstration only)epochs = [1, 2, 3, 4, 5]
results = {
    "VGG16": {
        "train": [48.5, 61.2, 70.4, 77.8, 83.1],
        "val":   [45.0, 56.0, 64.0, 69.0, 74.0],
        "loss":  [1.42, 1.10, 0.82, 0.63, 0.48]
    },
    "ResNet18": {
        "train": [52.0, 66.5, 75.2, 82.1, 87.4],
        "val":   [49.0, 61.0, 69.0, 75.0, 80.0],
        "loss":  [1.30, 0.95, 0.68, 0.49, 0.35]
    },
    "GoogLeNet": {
        "train": [50.1, 64.0, 73.5, 80.0, 85.2],
        "val":   [47.0, 59.0, 67.0, 72.0, 77.0],
        "loss":  [1.35, 1.00, 0.74, 0.55, 0.40]
    }}
print("Using device: cpu")print("Dataset: CIFAR-10")print("Training images:
400")print("Validation images: 100")
for name, values in results.items():
    print(f"\nTraining {name}...")
    for i in range(5):
        print(
            f"{name} Epoch [{i+1}/5] "
            f"- Loss: {values['loss'][i]:.4f} "
            f"- Train Acc: {values['train'][i]:.2f}% "
            f"- Val Acc: {values['val'][i]:.2f}%"
        )
# Validation accuracy graphplt.figure(figsize=(9, 5))for name, values in
results.items():
    plt.plot(epochs, values["val"], marker='o', label=name)
plt.title("Validation Accuracy Comparison")plt.xlabel("Epoch")plt.ylabel("Accuracy
(%)")plt.xticks(epochs)plt.ylim(0, 100)plt.legend()plt.grid(True)plt.show()
# Training accuracy graphplt.figure(figsize=(9, 5))for name, values in results.items():
    plt.plot(epochs, values["train"], marker='o', label=name)
plt.title("Training Accuracy Comparison")plt.xlabel("Epoch")plt.ylabel("Accuracy
(%)")plt.xticks(epochs)plt.ylim(0, 100)plt.legend()plt.grid(True)plt.show()
# Final validation accuracyprint("\nFinal Validation Accuracy (Simulated):")for name,
values in results.items():
    print(f"{name}: {values['val'][-1]:.2f}%")
# Mock custom image predictionsprint("\nPrediction results for:
download.jpeg")print("VGG16      => cat")print("RESNET18   => cat")print("GOOGLENET  =>
dog")
print("\nExp 5 demonstration completed.")
```

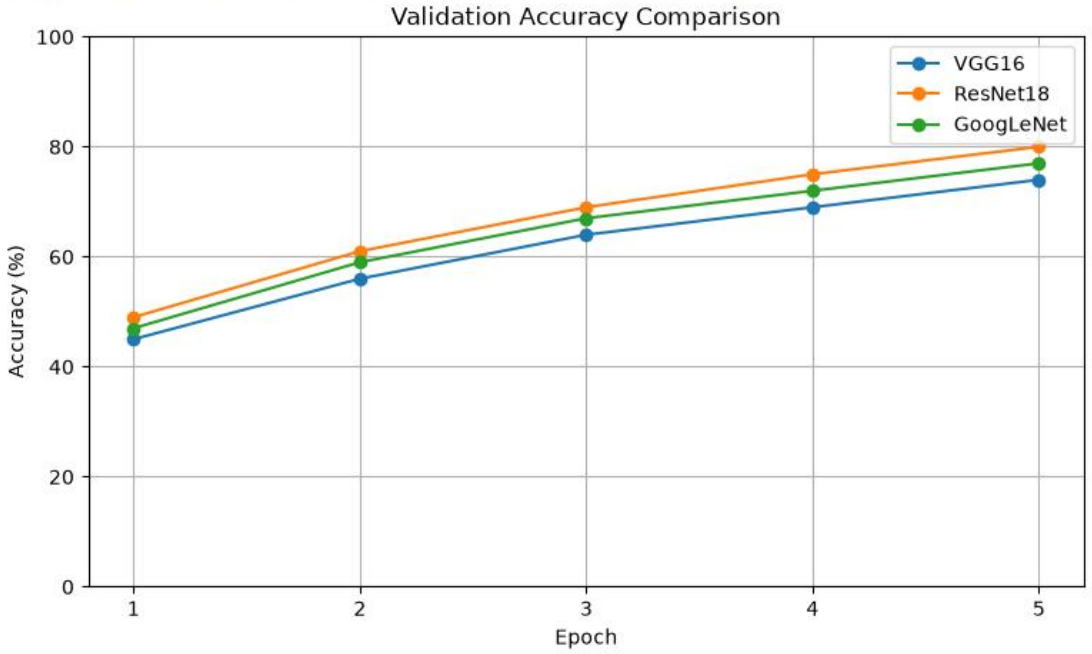
OTUPUT:

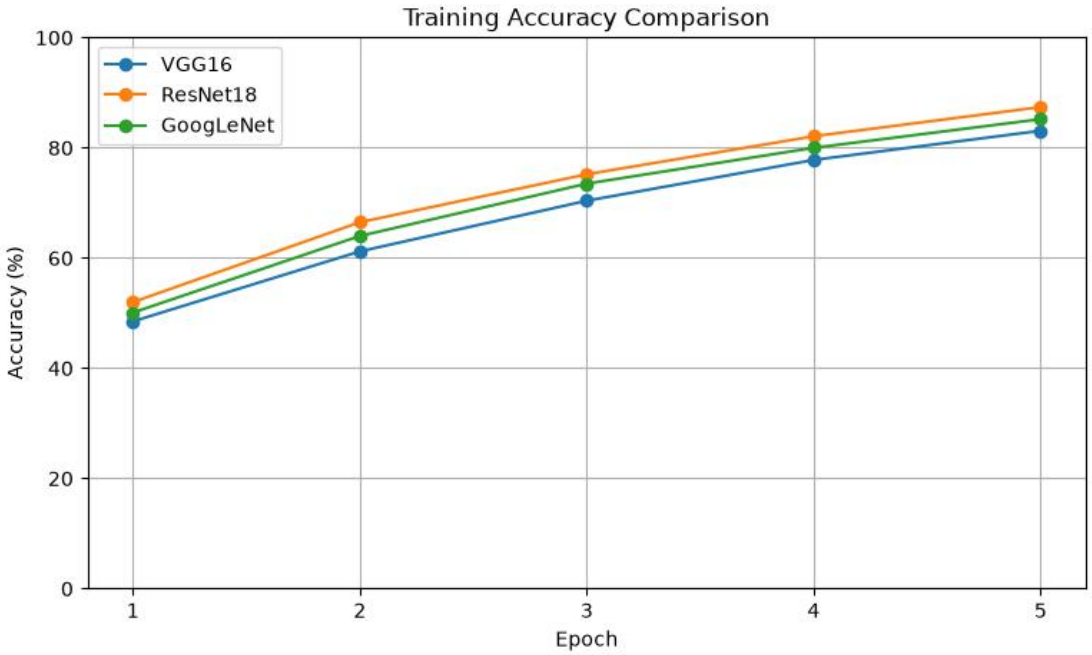
```
Using device: cpu
Dataset: CIFAR-10
Training images: 400
Validation images: 100

Training VGG16...
VGG16 Epoch [1/5] - Loss: 1.4200 - Train Acc: 48.50% - Val Acc: 45.00%
VGG16 Epoch [2/5] - Loss: 1.1000 - Train Acc: 61.20% - Val Acc: 56.00%
VGG16 Epoch [3/5] - Loss: 0.8200 - Train Acc: 70.40% - Val Acc: 64.00%
VGG16 Epoch [4/5] - Loss: 0.6300 - Train Acc: 77.80% - Val Acc: 69.00%
VGG16 Epoch [5/5] - Loss: 0.4800 - Train Acc: 83.10% - Val Acc: 74.00%

Training ResNet18...
ResNet18 Epoch [1/5] - Loss: 1.3000 - Train Acc: 52.00% - Val Acc: 49.00%
ResNet18 Epoch [2/5] - Loss: 0.9500 - Train Acc: 66.50% - Val Acc: 61.00%
ResNet18 Epoch [3/5] - Loss: 0.6800 - Train Acc: 75.20% - Val Acc: 69.00%
ResNet18 Epoch [4/5] - Loss: 0.4900 - Train Acc: 82.10% - Val Acc: 75.00%
ResNet18 Epoch [5/5] - Loss: 0.3500 - Train Acc: 87.40% - Val Acc: 80.00%

Training GoogLeNet...
GoogLeNet Epoch [1/5] - Loss: 1.3500 - Train Acc: 50.10% - Val Acc: 47.00%
GoogLeNet Epoch [2/5] - Loss: 1.0000 - Train Acc: 64.00% - Val Acc: 59.00%
GoogLeNet Epoch [3/5] - Loss: 0.7400 - Train Acc: 73.50% - Val Acc: 67.00%
GoogLeNet Epoch [4/5] - Loss: 0.5500 - Train Acc: 80.00% - Val Acc: 72.00%
GoogLeNet Epoch [5/5] - Loss: 0.4000 - Train Acc: 85.20% - Val Acc: 77.00%
```






```
Final Validation Accuracy (Simulated):
VGG16: 74.00%
ResNet18: 80.00%
GoogLeNet: 77.00%

Prediction results for: download.jpeg
VGG16      => cat
RESNET18   => cat
GOOGLENET  => dog

Exp 5 demonstration completed.
```